

Bypassing CSP using polyglot JPEGs

Bypassing CSP using polyglot JPEGs - Gareth Heyes, PortSwigger Research.

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Content

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[image: Gareth Heyes]

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[image: Bypassing CSP using polyglot JPEGs]

[James](#) challenged me to see if it was possible to create a polyglot JavaScript/JPEG. Doing so would allow me to bypass [CSP](#) on almost any website that hosts user-uploaded images on the same domain. I gleefully took up the challenge and begun dissecting the format. The first four bytes are a valid non-ASCII JavaScript variable 0xFF 0xD8 0xFF 0xE0. Then the next two bytes specify the length of the JPEG header. If we make that length of the header 0x2F2A using the bytes 0x2F 0x2A as you might guess we have a non-ASCII variable followed by a multi-line JavaScript comment. We then have to pad out the JPEG header to the length of 0x2F2A with nulls. Here's what it looks like:


```
FF D8 FF E0 09 3A 4A 46 49 46 2F 2A
```

Now we continue the rest of the JPEG header then pad with NULLs and inject our JavaScript payload:

```
FF D8 FF E0 09 3A 4A 46 49 46 2F 2A 01 01 00 48 00 48 00 00 00 00 00 00 00 ... (padding  
more nulls) 2A 2F 3D 61 6C 65 72 74 28 22 42 75 72 70 20 72 6F 63 6B 73 2E 22 29 3B 2F  
2A
```

Here is the smaller graphic:

[Polyglot JPEG smaller](#)

Impact

If you allow users to upload JPEGs, these uploads are on the same domain as your app, and your CSP allows script from "self", you can bypass the CSP using a polyglot JPEG by injecting a script and pointing it to that image.

Conclusion

In conclusion if you allow JPEG uploads on your site or indeed any type of file, it's worth placing these assets on a separate domain. When validating a JPEG, you should rewrite the JPEG header to ensure no code is sneaked in there and remove all JPEG comments. Obviously it's also essential that your CSP does not whitelist your image assets domain for script.

This post wouldn't be possible without the excellent work of [Ange Albertini](#). I used his [JPEG format graphic](#) extensively to create the polygot JPEG. [Jasvir Nagra](#) also inspired me with his blog post about [polyglot GIFs](#).

[PoC](#)

Update...

Mozilla are [fixing this in Firefox 51](#).

Visit our Web Security Academy to [learn more about cross-site scripting \(XSS\)](#)

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